Design and Evaluation of a 3D Cartographic Prototype for Visualizing Urban Thermal Stress at City-Block Scale

by YI-RU LIANG

Urban climate datasets are becoming increasingly complex, with 3D spatiotemporal simulations providing detailed information about changing environmental conditions. However, effective communication and exploration of these volumetric datasets remains a challenge in geovisualization due to visual occlusion and cognitive loads associated with 3D spatial environments. This thesis addresses this challenge by designing and evaluating a 3D cartographic prototype that visualizes Universal Thermal Climate Index (UTCI) values for the general public.

DATA

A pre-computed UTCI dataset, provided by Dr. Milena Vuckovic, contains ~1.9 million discrete points at a spatial resolution of 1×1×2 m across six height levels, representing pedestrian exposure up to average tree canopy height.

To examine the potential effect of urban trees on outdoor thermal conditions, the dataset contains four scenarios. These scenarios consist of an extreme warm day and an extreme cold day, with each condition simulated both with and without trees.

PROTOTYPE

The prototype was developed using Unity, with the C# visual analytics pipeline. It provides an interactive 3D environment in which users can explore UTCI data through navigation, interface controls, and spatial representations.

3D Visual Encodings

– Point clouds: Discrete 3D spheres positioned at exact simulation coordinates (Fig. 1(a)). [1]

– Horizontal planes: Continuous surfaces stacked from 0 to 10 meters at 2-meter vertical intervals (Fig. 2(b)). [1]

Interface Controls

An hourly time slider (1–24h), scenario toggles for *Tree Condition* and *Warm/Cold Condition*, and two independent opacity sliders to adjust the opacity of the UTCI visualization and the 3D building models separately (Fig. 2(a)).

Activity Mode

Translates thermal stress data into relatable everyday activities. When an Activity Mode is enabled, the visual output dynamically filters the active dataset to show only relevant areas:

– Walk the Dog: Isolates ground-level thermal safety areas for dogs. (Fig. 2(b))

– Sunbathing / Winter Walk: Isolates comfortable areas for solar exposure.

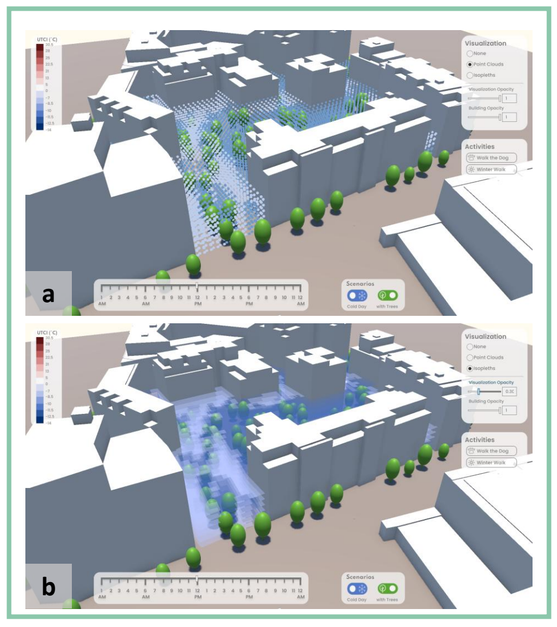
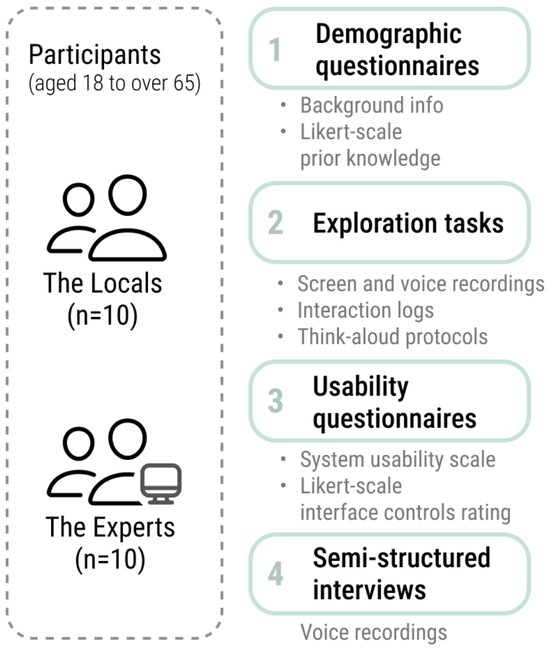


Fig. 1: 3D visual encodings in the prototype (a) point clouds and (b) horizontal planes with transparency

USER STUDY

To evaluate the usability and user experience, a mixed methods approach was used to collect quantitative measurements and qualitative data.



RESULTS

The prototype achieved a mean System Usability Scale score of 79.7, indicating an acceptable level of usability. [2]

Visual Encodings Trade-offs

– Point clouds were preferred when users navigated down to the pedestrian eye-level, as the gaps between points preserved spatial context and highlighted local variations.

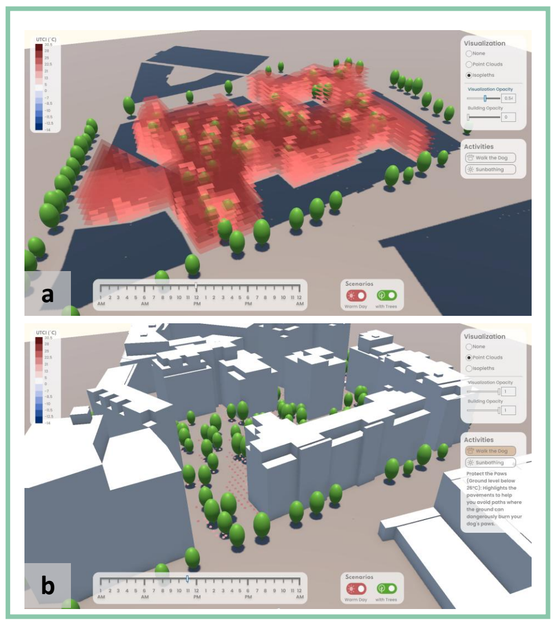


Fig. 2: Interface controls support visual analytics: (a) occlusion mitigation (b) Activity Mode - Walk the dog

– Horizontal planes were preferred by 70% of participants for overall task completion because they enabled rapid identification of thermal stress distribution from and better overview legibility from above.

Occlusion & Transparency

Transparency sliders mitigated visual occlusions but introduced uncertainty by altering color appearance, making it harder to match thermal stress values with the UTCI legend.

3D Familiarity vs. Experience

3D navigation familiarity was associated with more extensive exploration (more clicks, broader temporal coverage). 3D software experience was associated with higher task efficiency, particularly for complex analytical tasks.

User Frustration

Main sources of user frustration include occlusions caused by the visualizations, first-person flying navigation with unfamiliar keyboard controls, and ambiguous visual logic in the activity-based settings.

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